

Play-4: a longer bird-inspired call

Prepared for Henrik · Luftfugl motor firmware · 6 September 2026

Implemented and ready to listen

Play-4 is a new debug-only listening command. It retains the two successful piezo bands, 1950–2050 Hz and 4250–4350 Hz, and the clean, full-scale sine drive. One bout now lasts approximately five seconds, around eleven times play-3's 444 ms. Play-3, play-2 and routine station calls retain their existing compositions.

Type **buzzer play-4 3** to hear three complete bouts. **buzzer off** stops playback. Count accepts 1–200. The first bout after reset lasts **4.949 seconds**; three repetitions end after **16.949 seconds**. These are generated digital timings, not measured acoustic durations.

The sequence you will hear

Three phrases begin at 0, 2 and 4 seconds. Every phrase opens with two high-band notes: 100 ms sound, 100 ms silence, 100 ms sound, then 90 ms silence. This introduction is followed by the cluster below. Thus the middle phrase starts high before its low cluster.

Phrase	Cluster notes	Frequency	Sound/note	Gap/note
1	6	4250–4350 Hz	45–55 ms	30–40 ms
2	7	1950–2050 Hz	45–55 ms	35–45 ms
3	8	4250–4350 Hz	40–50 ms	25–35 ms

There are 27 notes altogether: six introductions and 21 shorter notes. Each note retains two contiguous halves with independent ± 25 Hz pitch draws, clamped to its band. There is no silent break between the halves. Note and gap durations vary within the table's bounds.

After phrases 1 and 2, silence fills the remaining time to the next two-second onset. The last note ends the bout. Another bout begins at six seconds, leaving approximately one second of silence. Each command generates a new pattern, then repeats that exact pattern for the requested count.

The third cluster contains more notes with shorter spacing; this is a chosen audible progression, not a scientifically established “maximum threat” formula.

What the bird research supports

Scope: original acoustic measurements and playback experiments on small passerines, with predator-response studies used to assess the “scare away” interpretation. No target species or intruder was selected; the earlier predator example is the working context. Frequencies are constrained by the user's bench measurements, not by a claim that these match a particular bird.

Notes last tens of milliseconds

Salis and colleagues measured great-tit F notes at 110 ± 40 ms, D notes at 50 ± 10 ms, F-F silence at 100 ± 30 ms, F-D at 90 ± 40 ms and D-D at 30 ± 6 ms (mean $\pm$ SD). Their synthetic template used two F notes followed by six D notes; that count was a playback choice. Table A1 gives D-D spacing as 0.03 s, while the methods say 0.3 s. I use the table, supported by the independent measurements below. [Salis et al., 2024 — acoustic measurements, Table A1](#).

Shortest possible gaps are not universally most urgent

In great tits, sparrowhawk-directed calls averaged 531 ± 33 ms, 7.095 ± 0.391 notes and 41 ± 2 ms inter-note gaps; owl-directed calls averaged 419 ± 24 ms, 6.063 ± 0.371 notes and 34 ± 1 ms gaps (mean $\pm$ SE). Higher threat raised whole-call rate from 7.97 to 15.03 calls per bird per minute, but lengthened gaps inside calls. [Kalb, Anger & Randler, 2019 — predator-specific mobbing calls](#).

Black-capped chickadees showed a different local timing change: greater threat shortened the first D note and the first-to-second D gap, while increasing the D-note section. The disagreement is species-specific evidence, not a reason to impose one universal timing rule. [Templeton, Greene & Davis, 2005 — predator-size information](#).

Real organization includes pauses and meaningful ordering

Japanese tits produce D strings of seven to ten notes. Suzuki and colleagues measured 50–150 ms natural inter-note intervals and used 20 or 30 calls per minute in playbacks, described as within natural ranges. Reversing alert-recruitment order reduced the normal response. Play-4's two-second phrase onset borrows the faster playback rate; it exceeds the great-tit mean above and is an engineering compromise, not a measured universal rate. [Suzuki, Wheatcroft & Griesser, 2016 — call ordering experiments](#).

Rhythm is only part of recognition

Salis et al. found mobbing responses in 3% of birds hearing a pure-tone control, versus 33% for complete artificial calls and 36% for natural calls. Correct temporal organization alone was insufficient. Our narrow-band clean sine is therefore a timing-inspired approximation, not a faithful great-tit vocalization. [Salis et al., 2024 — playback results](#).

What is chosen, and what is verified

Deliberate engineering choices

The three-phrase bout, high/low/high cluster arrangement, uniform integer timing variation, denser ending and exact repeated composition are sound-design decisions. Published measurements guide note durations and gaps; no paper establishes this exact recipe as the best deterrent. Introduction and cluster amplitude remain equal and full scale. No harmonics, clipping or amplitude envelope were added. The 100 kHz PWM/DDS transport remains a clean sine carrier implementation.

Possible bout length is **4885–5035 ms**. Phrase 1 occupies 810–920 ms, followed by 1080–1190 ms of silence. Phrase 2 occupies 915–1045 ms, followed by 955–1085 ms of silence. Phrase 3 occupies 885–1035 ms. Its following pause, only when repeating, is 965–1115 ms. Actual sound occupies 1505–1715 ms per bout.

The first generated example has phrase ends at 870, 2988 and 4949 ms; pauses are 1130, 1012 and 1051 ms. Later command invocations can differ within the stated bounds. The older 650 ms cap belongs to the original software generator; this independent player streams the complete longer pattern.

Scaring an intruder remains a separate test

Recorded mobbing calls produced stress-related behavioral changes in 15 captive raptors of seven species; this was not a field test of departure or of synthetic piezo tones. [Consla & Mumme, 2012 — captive raptor playback](#).

In natural powerful-owl observations, owls left their roost during 20% of 35 mobbing bouts. Live birds and their behavior were present, so audio alone was not isolated. [Pavey & Smyth, 1998 — natural owl mobbing](#).

Loudness, bird recognition and intruder deterrence have not been verified for play-4. Listening on the actual piezo is the next decision point. The earlier claim that two bands sound loud and clear comes from the user's bench listening, not an acoustic measurement made during this implementation.

Firmware verification completed

1000 compositions passed timing, pitch and phase-structure checks. Actual PWM samples repeated identically for counts 1, 2 and 200, with valid output levels and terminal silence. Existing DDS, play-3 and fixed-tone comparison tests also passed under their host checks. New tests used -Wall -Wextra -Werror and address/undefined-behavior sanitizers.

Debug and release builds passed with -Wall -Wextra and no compiler warnings. Play-3's four files are byte-identical to the pre-change workspace. Original generators and existing parsing/dispatch remain unchanged; only play-4 lifecycle hooks were added to the shared buzzer module. The release binary is byte-identical to the pre-play-3 production reference.

OpenOCD identified 4096 KiB flash and reported Verified OK. Console checks passed help, invalid counts, normal completion, count 200 with stop, and switching between play-4 and play-2/play-3. Tick overruns were zero; the controller reported IDLE at position 6, duty 0. No motion commands were issued. The console menu was reset after testing.

Source notes

Primary publications underpin the biological claims. All were consulted on 6 September 2026. The firmware measurements and bench constraints have separate provenance: source code/host tests/console readback for digital behavior, and the user's observations for piezo response.

Salis, A.; Badaire, F.; Coye, C.; Leroux, M.; Lengagne, T.; Schlenker, P.; Chemla, E. (October 2024). "Mechanisms of mobbing call recognition: exploring featural decoding in great tits." *Animal Behaviour* 216, 63–71. [Publisher article and DOI](#). Indexed full article, Appendix Table A1 and results inspected; direct publisher access intermittently returned 403. The 0.03/0.3 s discrepancy remains unresolved in the publication and is disclosed on page 2.

Kalb, N.; Anger, F.; Randler, C. (25 April 2019). "Subtle variations in mobbing calls are predator-specific in great tits (*Parus major*)." *Scientific Reports* 9, 6572. [Publisher article and DOI](#). Full publication also inspected through the [German National Library copy](#); quantitative results on page 2 independently checked.

Templeton, C. N.; Greene, E.; Davis, K. (24 June 2005). "Allometry of alarm calls: black-capped chickadees encode information about predator size." *Science* 308, 1934–1937. [Publication DOI](#). Primary paper read through the [author-hosted PDF](#). Used for the direction of temporal changes, not an invented millisecond estimate.

Suzuki, T. N.; Wheatcroft, D.; Griesser, M. (8 March 2016). "Experimental evidence for compositional syntax in bird calls." *Nature Communications* 7, 10986. [Open-access article](#). Full methods and experimental context inspected. Playback rates are distinguished from measured natural inter-note intervals.

Consla, D. J.; Mumme, R. L. (21 August 2012, first publication). "Response of Captive Raptors to Avian Mobbing Calls: the Roles of Mobber Size and Raptor Experience." *Ethology* 118, 1063–1071. [Publisher abstract](#). Abstract-level access; no claim about uninspected detailed results or optimal temporal parameters.

Pavey, C. R.; Smyth, A. K. (February 1998). "Effects of avian mobbing on roost use and diet of powerful owls, *Ninox strenua*." *Animal Behaviour* 55, 313–318. [Publication DOI](#). [PubMed abstract](#) inspected; natural observation, not an audio-only experiment.

Further searching stopped after the timing choices had direct support, the contradictory urgency patterns were explained by species, and the recognition/deterrence limits were established. There is no unresolved evidence gap that could justify claiming this narrow-band synthetic sequence is a validated intruder deterrent.